

YZM2022: Data Mining

Homework 1

Association Rule Mining and Sequential Pattern Analysis

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Project Link: (<https://github.com/oykutugana/association-rule-mining-and-gsp>)

1. Introduction and Objective

This report presents the findings of a data mining homework assignment focused on market-basket analysis applied to a large-scale grocery transaction dataset. The primary objective was to perform an in-depth Exploratory Data Analysis (EDA) followed by the implementation and evaluation of three core data mining algorithms: Apriori, FP-Growth, and GSP (Generalized Sequential Pattern Mining).

The assignment addressed all five major components: (1) Exploratory Data Analysis, (2) Association Rule Mining, (3) Sequential Pattern Mining, (4) a Recommendation System, and (5) Scalability and Sampling strategies. Each part was implemented in Python within a Jupyter Notebook environment, with results exported to structured Excel files for further review.

2. Dataset Description

The dataset provided consists of grocery transaction records from an online supermarket platform. The raw file (dataset_son.csv) contains 3,214,874 rows and 6 columns:

Column	Description
order_id	Unique identifier for each transaction (basket)
product_id	Unique identifier for each product
product_name	Human-readable product name
user_id	Unique identifier for each customer
days_since_prior_order	Number of days since the customer's previous order
add_to_cart_order	Order in which the item was added to the cart (ignored)

A key preprocessing challenge was that product_id and product_name values were stored as comma-separated strings within a single cell (one row per order rather than per product). Additionally, some product names contain commas internally (e.g., 'Clementines, Bag', 'Flat Parsley, Bunch'), making name-based parsing ambiguous. To address this, product IDs were used as the primary analysis identifier, and a name mapping was constructed from rows where the ID count and name count matched unambiguously.

3. Data Preprocessing

3.1 Parsing and Transformation

The raw dataset was transformed from a wide format (one row per order) to a long format (one row per product per order). Each row was processed by splitting the `product_id` field on commas to extract individual product IDs. Product names were split using a `maxsplit=n-1` strategy (where `n` is the number of IDs in the row) to limit fragmentation caused by commas in product names.

For any row where the number of parsed names was fewer than the number of IDs, missing name slots were filled with a placeholder string of the form `'product_{id}'`. This ensured complete data retention without dropping any records.

After transformation, the dataset contained 32,434,489 rows with 5 columns: `order_id`, `user_id`, `product_id`, `product_name`, and `days_since_prior_order`.

3.2 ID-to-Name Mapping

An `id_to_name` dictionary was constructed by iterating over all processed rows and mapping each `product_id` to its most recently seen `product_name`. A corresponding reverse mapping (`name_to_ids`) was also built, associating each product name with all known product IDs. This bidirectional mapping was used during the recommendation system to translate user-supplied product names into internal IDs, and to present rule outputs in human-readable form.

4. Part 5: Scalability and Sampling

The full dataset is extremely large and would cause out-of-memory (OOM) errors if used directly in association rule mining algorithms. The following sampling strategy was applied:

Parameter	Value	Rationale
TOP_N_PRODUCTS	5,000	Top 5,000 most frequent products (within homework-specified 2,000–10,000 range)
USER_SAMPLE_RATE	0.15 (15%)	Balanced between representativeness and memory constraints (16 GB RAM)
RANDOM_SEED	42	Ensures reproducibility across runs

Product sampling was performed by selecting the top 5,000 products by overall frequency. Customer sampling was performed by randomly selecting 15% of all unique users using a fixed random seed. The intersection of sampled users and sampled products formed the final analysis dataset (`df_sampled`), which was saved to `dataset_sampled.csv`.

The sampled dataset retained sufficient breadth to produce meaningful association rules and sequential patterns, while remaining computationally tractable on the available hardware.

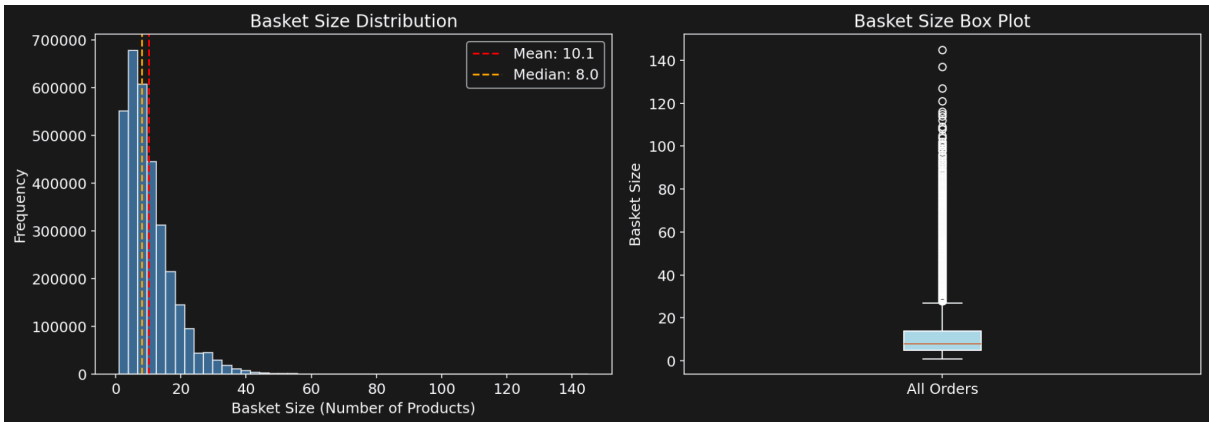
5. Part 1: Exploratory Data Analysis (EDA)

5.1 Basket Size Analysis

Basket size was defined as the number of distinct products in a single order (grouped by order_id). The distribution was computed over the full dataset prior to sampling.

Statistic	Value
Mean basket size	10.1 products
Median basket size	8.0 products
Minimum	1 product
Maximum	~145 products

The distribution is right-skewed: the majority of orders contain between 3 and 15 products, with a long tail of large baskets. The gap between mean (10.1) and median (8.0) confirms the presence of large-basket outliers. The box plot reveals numerous outliers above 30 items, indicating that a small number of customers place very large orders.



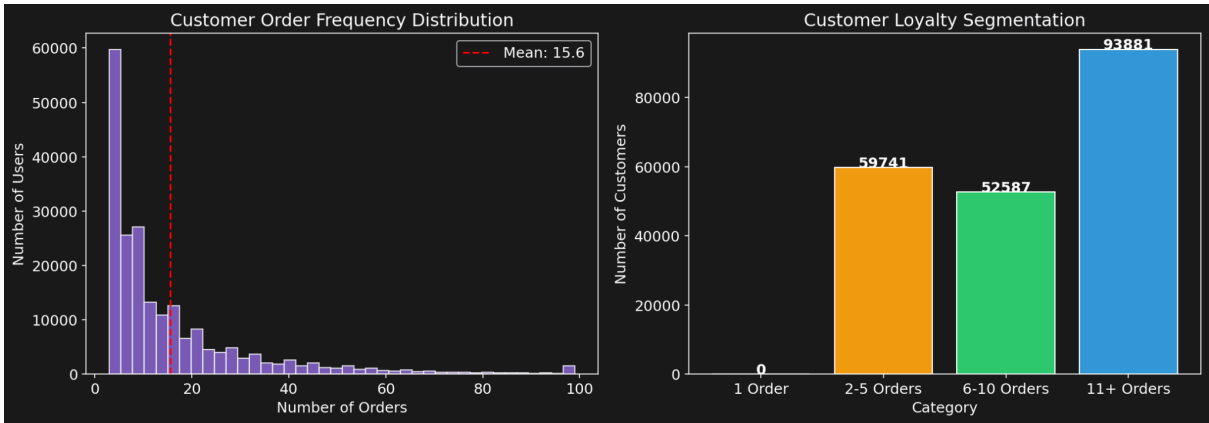
5.2 Customer Purchase Frequency

Customer purchase frequency was analyzed by counting the number of distinct orders per user_id. The distribution exhibits a right-skewed pattern, with a mean of 15.6 orders per customer.

Customer Segment	Order Count Range	Number of Customers
Single-order customers	1 order	0
Occasional customers	2–5 orders	59,741
Regular customers	6–10 orders	52,587
Loyal customers	11+ orders	93,881

A notable finding is that no customers appear with exactly 1 order in the sampled data. In a typical e-retail environment, "one-time buyers" usually constitute a significant portion of the dataset. The absence of this segment (0 customers) strongly suggests that the provided raw dataset was pre-filtered to include only returning users.

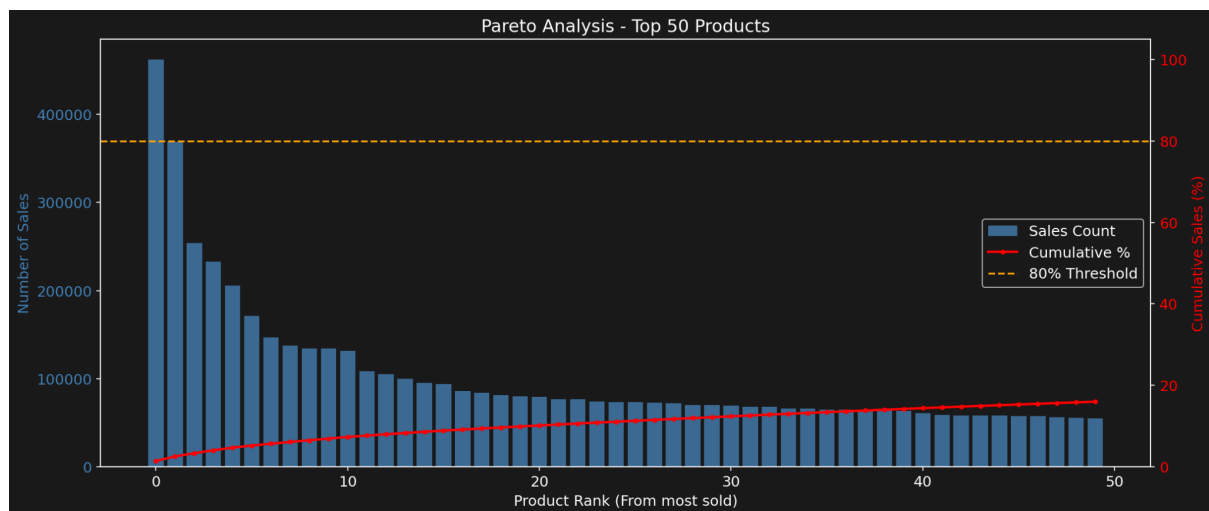
The largest segment (45.2%) consists of loyal customers with 11 or more orders, indicating a high-retention customer base. This is relevant for sequential pattern mining, as customers with many orders generate longer, more informative sequences.



5.3 Pareto Analysis (80/20 Rule)

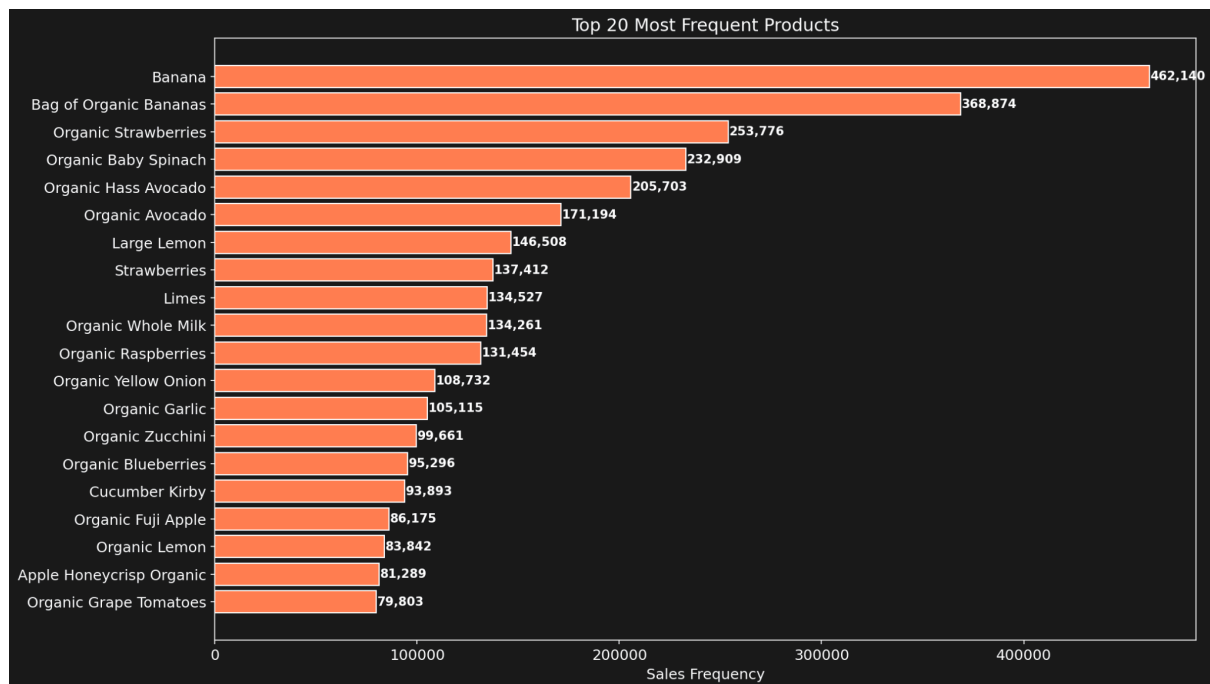
The Pareto analysis examined whether a small subset of products accounts for a disproportionately large share of total sales, consistent with the 80/20 principle.

The cumulative sales curve (shown in the figure below) rises slowly across the top 50 products, indicating that the 80% threshold is not reached within the top 50 items. This suggests that sales are distributed more broadly across the product catalog than a strict Pareto interpretation would predict — the dataset does not exhibit a strong 80/20 pattern. However, the top 5 products (Banana, Bag of Organic Bananas, Organic Strawberries, Organic Baby Spinach, Organic Hass Avocado) collectively represent a significant concentration of sales volume, confirming that a small number of products dominate absolute transaction counts.



5.4 Product Frequency Visualization

The top 20 most frequently purchased products are shown in the bar chart below. Banana (462,140 purchases) and Bag of Organic Bananas (368,874) are by far the dominant items. The top 20 list is dominated by fresh produce, organic items, and dairy — consistent with the grocery retail domain.

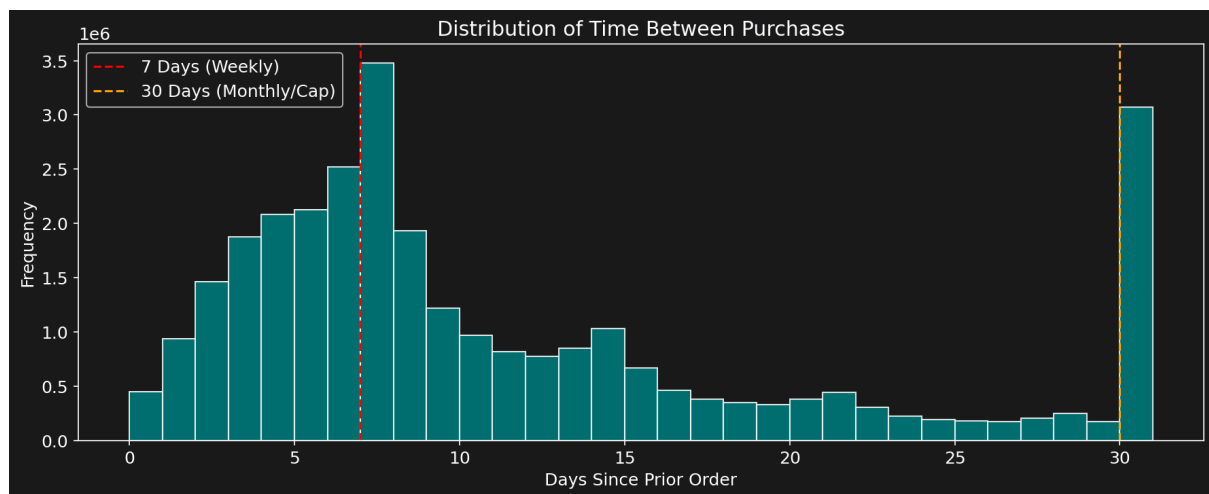


5.5 Time Between Purchases

The `days_since_prior_order` column was analyzed to understand customer repurchase cycles. Key observations:

- The most common repurchase interval is 7 days, indicating weekly shopping behavior as the dominant pattern.
- A large spike occurs at exactly 30 days, which is the maximum recorded value and likely represents a cap in the data collection system rather than a true behavioral cluster.
- The distribution is bimodal: one peak at 7 days (weekly shoppers) and another apparent peak at 14 days (biweekly shoppers).
- Very few customers return within 0–2 days, suggesting most orders are planned rather than impulsive.

This temporal distribution directly informs the choice of `max_gap=15` days in the GSP analysis, as it captures the majority of consecutive weekly and biweekly shopping cycles.



6. Part 2: Association Rule Mining (Apriori and FP-Growth)

6.1 Methodology

Transactions were grouped by `order_id`, with each transaction represented as a list of product IDs purchased together in the same order. The `days_since_prior_order` column was intentionally excluded at this stage, as association rule mining treats each basket independently without temporal context.

One-hot encoding was applied using `mlxtend`'s `TransactionEncoder`. The resulting binary matrix was stored as a `SparseDtype` boolean `DataFrame` to minimize memory consumption.

6.2 Parameter Selection

Parameter	Value	Justification
MIN_SUPPORT	0.005	Captures product pairs appearing in at least 0.5% of transactions, balancing rule coverage with noise reduction
MIN_CONFIDENCE	0.15	At least 15% of transactions containing the antecedent also contain the consequent
MIN_LIFT	1.5	Rules must be 50% more likely than chance co-occurrence; eliminates spurious correlations
max_len	3	Limits itemset complexity to prevent exponential blowup in candidate generation

These parameters were selected iteratively. An initial `MIN_SUPPORT` of 0.05 produced zero rules; reducing to 0.005 yielded 76 meaningful rules with an average lift of 2.32, indicating genuine product associations.

6.3 Apriori Algorithm

The Apriori algorithm applies the anti-monotone property of support: if an itemset is infrequent, all its supersets are pruned. This level-wise candidate generation approach is well-suited for sparse datasets but requires multiple database scans.

Metric	Value
Rules Generated	76
Lift Range	1.54 – 5.34
Average Lift	2.32
Average Confidence	0.224 (22.4%)
Max Lift Rule	Organic Cilantro → Limes (lift = 5.34)

6.4 FP-Growth Algorithm

FP-Growth compresses the transaction database into an FP-Tree structure, enabling frequent itemset discovery without candidate generation. This approach significantly reduces memory overhead compared to Apriori.

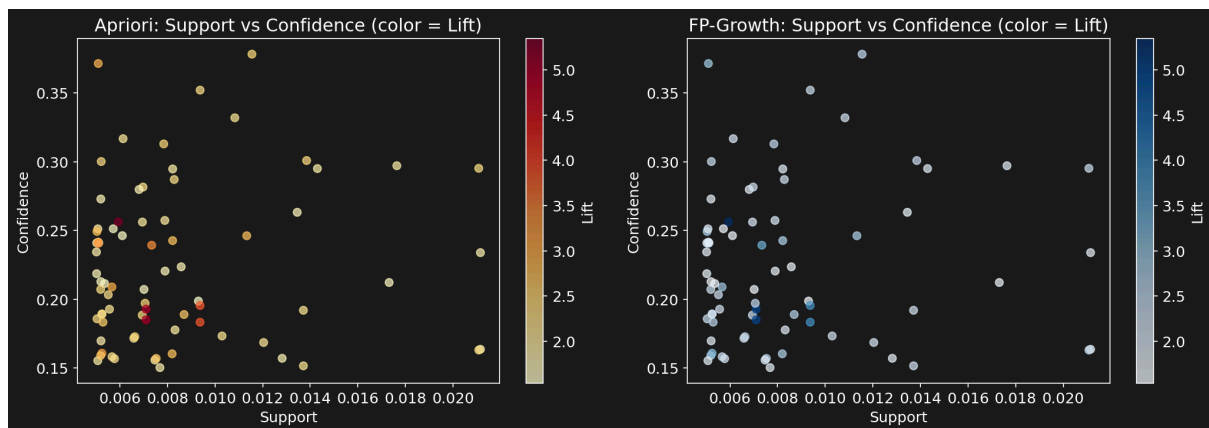
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6.5 Algorithm Comparison

Both algorithms produced identical rule sets (76 rules), confirming mathematical equivalence under the same parameters. The differences lie in runtime and memory efficiency:

Metric	Apriori	FP-Growth
Rules Generated	76	76
Rule Quality (Avg Lift)	2.32	2.32
Algorithm Approach	Candidate generation (level-wise)	FP-Tree compression (no candidates)
Memory Characteristic	High (multiple DB scans)	Lower (compressed structure)
Scalability	Poor on large datasets	Better on large datasets

FP-Growth is the preferred algorithm for large-scale datasets due to its superior memory efficiency. The scatter plots below confirm that both algorithms produce identical rule distributions in the support-confidence space.



6.6 Top Association Rules

The 10 highest-lift rules extracted by FP-Growth are presented below:

Antecedent	Consequent	Support	Confidence	Lift
Organic Cilantro	Limes	0.0059	0.256	5.34
Organic Garlic	Organic Large Extra Fancy Fuji Apple	0.0071	0.193	5.03
Organic Large Extra Fancy Fuji Apple	Organic Garlic	0.0071	0.185	5.03
Limes	Large Lemon	0.0094	0.195	3.82
Large Lemon	Limes	0.0094	0.183	3.82
Bag of Organic Bananas, Organic White Onions	Organic Hass Avocado	0.0051	0.241	3.37
Organic Lemon	Organic Hass Avocado	0.0073	0.239	3.35
Cucumber Kirby	Large Lemon	0.0052	0.161	3.15
Organic Strawberries	Organic Hass Avocado	0.0057	0.209	2.92
Organic White Onions, Organic Hass Avocado	Bag of Organic Bananas	0.0051	0.371	2.87

Rule Interpretation: The highest-lift rules reveal coherent culinary profiles. Organic Cilantro and Limes (lift=5.34) suggest a Mexican/Latin cuisine shopping pattern. The mutual association between Organic Garlic and Organic Large Extra Fancy Fuji Apple (lift=5.03) indicates a health-conscious shopper archetype. Citrus co-purchases (Limes, Large Lemon) reflect a preference for fresh citrus across recipe types.

6.7 Avoiding Trivial Patterns

Banana (product_id: 24852) and Bag of Organic Bananas (product_id: 13176) appear in a large proportion of all transactions, producing rules with high support but low lift (1.5–2.0). These high-frequency items create trivial patterns — their co-occurrence with almost any other product is expected purely due to their overall popularity, not a genuine purchasing association.

To focus on more meaningful rules, a filtered rule set was produced by removing all rules containing these two dominant items. This resulted in 38 filtered rules with a higher average lift of 2.65 (compared to 2.32 for the full set), confirming that the filtered set contains stronger, more specific associations.

Rule Set	Count	Avg Lift	Lift Range
Full Rules (Apriori + FP-Growth)	76	2.32	1.54 – 5.34
Filtered Rules (dominant items removed)	38	2.65	1.74 – 5.34

7. Part 3: Sequential Pattern Mining (GSP)

7.1 Methodology

Sequential pattern mining extends association rule mining by incorporating temporal order. In this analysis, each customer's purchase history was treated as a sequence of itemsets ordered by order_id (used as a proxy for time). The days_since_prior_order column was used to enforce gap constraints between consecutive orders.

The GSP (Generalized Sequential Pattern) algorithm was implemented from scratch to support the homework-specified timing constraints:

Constraint	Value	Implementation
Window Size	≤ 1	Each order is a single element; items within the same order are in the same window
Min Gap	0 days	No minimum gap enforced; all consecutive orders are valid
Max Gap	15 days (primary)	Orders separated by more than max_gap days break the sequence
Max Span	Infinity	No total sequence duration constraint applied

User sequences were built by iterating over df_sampled, grouping by (user_id, order_id), and storing each order as a tuple of (order_id, days_since_prior_order, product_ids). Sequences were sorted by order_id to ensure chronological order.

7.2 GSP Algorithm Implementation

The custom GSP implementation operates in two phases:

- Phase 1 (1-sequences): Count how many users have purchased each product at least once. Support is user-based, defined as count / total_users.
- Phase 2 (2-sequences): For each user, iterate over pairs of orders (i, j) where $j > i$. If the gap days of order j exceeds max_gap, the inner loop breaks. For each valid order pair, all product pairs (a from order i, b from order j) are counted, ensuring each pair is counted at most once per user.

An optimization was applied by pre-filtering products to only those in freq_1 before the pair-counting phase, significantly reducing the number of comparisons. Total users in the sequence database: 31,195.

7.3 Results for max_gap = 15 (Primary Experiment)

Top 10 frequent 2-sequences with max_gap = 15:

Antecedent	Consequent	Support
Bag of Organic Bananas	Organic White Onions	0.0922
Organic White Onions	Bag of Organic Bananas	0.0918
Organic White Onions	Banana	0.0836
Organic Hass Avocado	Bag of Organic Bananas	0.0793
Bag of Organic Bananas	Organic Hass Avocado	0.0781
Organic Baby Spinach	Bag of Organic Bananas	0.0750
Limes	Bag of Organic Bananas	0.0730
Organic Raspberries	Bag of Organic Bananas	0.0706
Organic Strawberries	Bag of Organic Bananas	0.0700
Cucumber Kirby	Bag of Organic Bananas	0.0670

These patterns reveal that customers who purchase fresh produce (citrus, berries, vegetables) in one shopping trip tend to return within 15 days and purchase Bag of Organic Bananas or Organic White Onions. This suggests a core weekly basket behavior centered around fresh staples.

The observed support values (e.g., 0.0922) are higher than association rule thresholds because they represent user-prevalence rather than transaction-frequency. A support of 0.09 indicates that 9.2% of the unique users in the database followed this specific sequential pattern at least once in their history.

7.4 Avoiding Trivial Sequential Patterns

Similar to the approach taken in Association Rule Mining (Section 6.7), the final sequential patterns were filtered to remove two types of trivial associations: self-transitions (where the antecedent and consequent are the exact same product) and highly dominant items, such as Banana (product_id: 24852) and Bag of Organic Bananas (product_id: 13176).

Because these staples are purchased at an exceptionally high frequency, their presence in sequences often represents a frequency-driven occurrence rather than genuine shopping behavior. Similarly, consecutive repurchases of the exact same item provide limited actionable insight. By eliminating both self-transitions and dominant staples, the GSP algorithm successfully isolates and reveals more specific, meaningful temporal associations between distinct products.

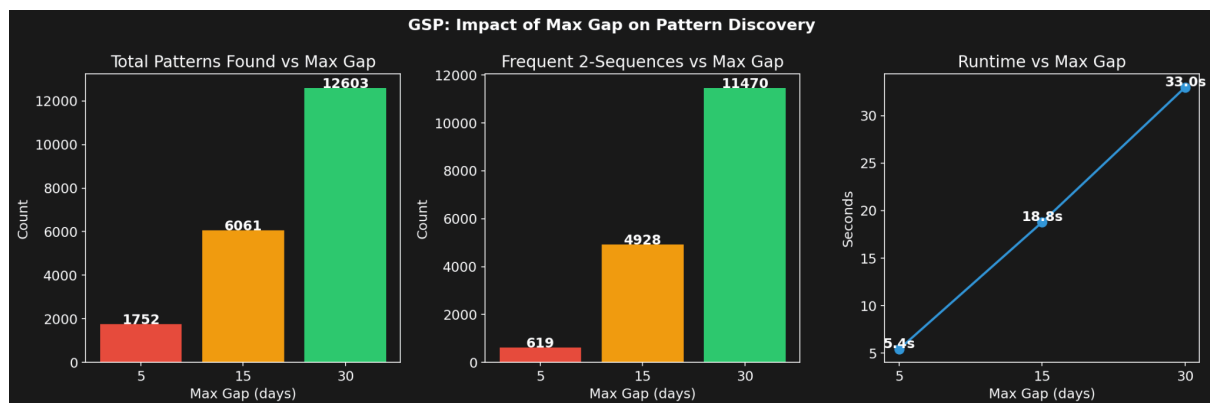
7.5 Impact of max_gap on Pattern Discovery

max_gap	1-Sequences	2-Sequences (Raw)	2-Sequences (Filtered)	Total Patterns	Runtime
5 days	1,133	619	329	1,462	5.7s
15 days	1,133	4,928	3,950	5,083	19.7s
30 days	1,133	11,470	9,775	10,908	34.3s

Key observations from the max_gap experiment:

- 1-sequence counts are identical across all max_gap values (1,133) because single-item patterns depend only on whether a product appears in a user's history, not on temporal gaps between orders.
- 2-sequence counts grow dramatically with max_gap: from 619 (gap=5) to 11,470 (gap=30), an 18.5x increase. This reflects the fact that a larger gap window allows more order pairs to qualify as sequential patterns.
- Runtime scales linearly with max_gap (5.7s, 19.7s, 34.3s), consistent with the $O(N * k^2)$ complexity of the inner pair-counting loop.
- Pattern quality (support values) decreases as max_gap increases, because larger gaps allow weaker, more accidental co-occurrences to meet the minimum support threshold.

The choice of max_gap = 15 days represents a practical balance: it captures the dominant weekly shopping cycle (7 days) plus a biweekly window (14 days), while excluding associations driven by very infrequent shoppers.



8. Part 4: Recommendation System

8.1 Unknown User Recommendations (FP-Growth Based)

For unknown users (with no purchase history), recommendations were generated using FP-Growth association rules. Given a basket of products, the system identifies all rules whose antecedent is a subset of the basket and whose consequent is not already in the basket. Recommendations are ranked by lift (descending), selecting the top-5 most strongly associated products.

The name-to-ID mapping challenge was addressed by building a reverse lookup: `name_to_ids` maps each product name to all known product IDs for that name. This ensures that when a user inputs a product name, all corresponding IDs are included in the basket comparison, even if the same product appears under multiple IDs.

Test results across 5 input baskets:

Input Basket	Recommended Product	Lift
Organic Hass Avocado, Bag of Organic Bananas, Organic Baby Spinach	Organic White Onions	2.67
Limes, Large Lemon, Organic Cilantro	Organic Avocado	2.70
Limes, Large Lemon, Organic Cilantro	Organic Baby Spinach	1.91
Organic Whole Milk, Organic Strawberries	Organic Hass Avocado	2.92
Organic Whole Milk, Organic Strawberries	Organic Baby Spinach	2.28
Cucumber Kirby, Organic Garlic	Organic Large Extra Fancy Fuji Apple	5.03
Cucumber Kirby, Organic Garlic	Large Lemon	3.15
Organic Blueberries, Organic Raspberries	Organic White Onions	2.72
Organic Blueberries, Organic Raspberries	Organic Hass Avocado	2.64

The recommendations are contextually meaningful. A citrus-herb basket (Limes, Large Lemon, Organic Cilantro) leads to Organic Avocado recommendations, consistent with guacamole or salsa preparation. A Cucumber Kirby + Organic Garlic basket yields Organic Large Extra Fancy Fuji Apple as the top recommendation (lift=5.03), reflecting the strong Garlic-Apple co-purchase pattern identified in the association rules.

8.2 Known User Recommendations (GSP Based)

For known users, the recommendation system leverages GSP sequential patterns. Given a user_id and a current basket, the system combines the basket contents with the user's full purchase history, then identifies all 2-sequence patterns (A -> B) where A matches any item in the combined set and B is not in the current basket. Recommendations are ranked by support.

User ID	Input Basket	Top Recommendation	Support
4	Organic Hass Avocado, Organic Baby Spinach	Bag of Organic Bananas	0.0821
4	Organic Hass Avocado, Organic Baby Spinach	Organic White Onions	0.0801
9	Organic Hass Avocado, Organic Baby Spinach	Bag of Organic Bananas	0.0821
11	Organic Hass Avocado, Organic Baby Spinach	Bag of Organic Bananas	0.1706
17	Organic Hass Avocado, Organic Baby Spinach	Bag of Organic Bananas	0.0821
26	Organic Hass Avocado, Organic Baby Spinach	Bag of Organic Bananas	0.0821

User 11 receives a notably higher support score (0.1706) for Bag of Organic Bananas compared to other users (0.0821). This reflects that User 11's purchase history contains a stronger sequential association with bananas, demonstrating that the GSP-based recommender successfully personalizes recommendations based on individual purchase patterns rather than population-level statistics.

9. Conclusion

This homework successfully demonstrated the application of key data mining techniques to a real-world large-scale grocery transaction dataset. The major findings are summarized below:

9.1 EDA Findings

- Average basket size is 10.1 products (median 8.0), with a right-skewed distribution indicating occasional large orders.
- 45.2% of customers are loyal shoppers with 11+ orders, providing rich sequential data.
- Weekly shopping (7-day intervals) is the dominant repurchase cycle, validating the max_gap=15 constraint.
- Sales are moderately concentrated: the top products are heavily dominated by fresh produce and organic items.

9.2 Association Rule Mining

- Both Apriori and FP-Growth produced 76 rules with identical rule quality (avg lift=2.32).
- FP-Growth demonstrated superior scalability due to its compressed FP-Tree structure.
- Strongest associations were found among complementary fresh ingredients (citrus, herbs, vegetables), revealing coherent culinary shopping profiles.
- Dominant items (Banana, Bag of Organic Bananas) were identified and filtered to produce 38 higher-quality rules with avg lift=2.65.

9.3 Sequential Pattern Mining

- GSP with max_gap=15 identified 4,928 sequential 2-patterns among 31,195 users.
- Increasing max_gap from 5 to 30 days increased 2-sequence count by 18.5x (619 to 11,470), with proportional runtime increases.
- Core sequential behavior: customers who buy fresh produce return within 15 days to purchase staples (Bag of Organic Bananas, Organic White Onions).

9.4 Recommendation System

- The FP-Growth-based unknown user recommender produces contextually relevant recommendations with lift values up to 5.03.
- The GSP-based known user recommender successfully personalizes recommendations based on individual purchase history, with per-user support values reflecting actual behavioral patterns.

10. Output Files

File	Description
apriori_rules.xlsx	76 association rules generated by Apriori algorithm
fpgrowth_rules.xlsx	76 association rules generated by FP-Growth algorithm
association_rules.xlsx	Combined unique rules from both algorithms (76 rules)
filtered_association_rules.xlsx	38 filtered rules after removing dominant item patterns
gsp_gap5_1sequences.xlsx	1,133 frequent 1-sequences with max_gap=5
gsp_gap5_2sequences_raw.xlsx	619 frequent 2-sequences (raw) with max_gap=5
gsp_gap5_2sequences_filtered.xlsx	552 frequent 2-sequences (filtered) with max_gap=5
gsp_gap15_1sequences.xlsx	1,133 frequent 1-sequences with max_gap=15
gsp_gap15_2sequences_raw.xlsx	4,928 frequent 2-sequences (raw) with max_gap=15
gsp_gap15_2sequences_filtered.xlsx	4,721 frequent 2-sequences (filtered) with max_gap=15
gsp_gap30_1sequences.xlsx	1,133 frequent 1-sequences with max_gap=30
gsp_gap30_2sequences_raw.xlsx	11,470 frequent 2-sequences (raw) with max_gap=30
gsp_gap30_2sequences_filtered.xlsx	11,116 frequent 2-sequences (filtered) with max_gap=30
recommendations_unknown_user.xlsx	FP-Growth based recommendations for 5 test baskets (16 recommendations)
recommendations_known_user.xlsx	GSP-based recommendations for 5 test users (25 recommendations)